How fast can a driven Markov network oscillate?

A four-state counterexample to the affine Karpelevich conjecture, and the exact dimension at which it fails

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Abstract

The eigenvalues of a continuous-time Markov rate matrix decide how fast its probability distribution relaxes and how strongly it can oscillate. For transition matrices of discrete-time chains, the exact set of possible eigenvalues is a classical object, the Karpelevich region. In 2025, Xu, Kolchinsky, Delvenne and Ito conjectured that the eigenvalues of rate matrices fill a "squashed" copy of this region: stretch it by the fastest escape rate, and flatten its imaginary direction by a factor set by the strongest thermodynamic driving force around any cycle (the largest cycle affinity per step). We show that this conjecture is false. An explicit four-state network with integer rates, about $8.4\,k_{\mathrm{B}}T$ of driving around its strongest cycle, has an eigenvalue outside the conjectured region; one integer inequality proves it. For networks of at most three states the conjectured containment is true, so four is the smallest failing size. The mechanism is simple: adding weak "resetting" transitions of size ε moves the eigenvalues only by an amount proportional to ε, but relaxes the affinity factor by an amount proportional to $\varepsilon^{2/3}$, which is much larger. This gives counterexamples near a whole boundary arc in four states, an explicit interval $0 < \varepsilon \le 1/100$, exactly certified counterexamples in five and six states, and an asymptotic construction in every dimension $n \ge 4$. All counterexamples use networks in which every pair of states is connected; whether the conjecture holds for single-cycle networks, the setting in which it was tested, remains open. The proved ellipse and polygon bounds are unaffected.

1 Introduction

1.1 Why the eigenvalue region matters

A finite-state Markov jump process is described by a rate matrix (generator) Q. A nonreal eigenvalue λ of Q is a mode that oscillates at angular frequency $|\operatorname{Im}\lambda|$ and decays at rate $-\operatorname{Re}\lambda$. The ratio $|\operatorname{Im}\lambda|/|\operatorname{Re}\lambda|$ measures how many coherent oscillations the mode completes before it decays. Such modes underlie biochemical clocks and molecular motors, and their coherence is limited by thermodynamics: a network at equilibrium has a real spectrum, and oscillations require driving. Barato and Seifert proposed a bound on coherence in terms of the driving force and the cycle length [1], later proved by Ohga, Ito and Kolchinsky [8]. Uhl and Seifert conjectured an ellipse that contains the whole rescaled spectrum [10]; Kolchinsky, Ohga and Ito proved it for discrete multicyclic generators and added a polygonal bound [7]. Bounds of this kind are also used in reverse: from a measured relaxation spectrum one infers a minimum dissipation.

The driving force is measured by cycle affinities. For a directed cycle c of a network, the affinity F_c is the logarithm of the product of forward rates divided by the product of backward rates. In isothermal systems it is the entropy produced, in units of k_{B}, each time the cycle is completed.

For discrete time the corresponding problem was solved long ago. The Karpelevich region Ω_n is the set of all eigenvalues of all $n \times n$ stochastic matrices [5, 3]. It is a curvilinear star inside the unit disc whose boundary arcs meet the unit circle at the roots of unity of order at most n (Figure 1).

1.2 The conjecture and the answer

Xu, Kolchinsky, Delvenne and Ito [12, Supplement, Eq. (S3)] proposed the natural continuous-time analogue: the eigenvalues of rate matrices fill an affine image of Ω_n, shifted and stretched by the largest escape rate α and flattened vertically by the factor $\beta' = \tanh(\max_c F_c/(2|c|))$ (the precise statement is quoted in Section 2). Undoing this transformation, every eigenvalue λ of Q should give a "rescaled eigenvalue"

$$\zeta = 1 + \frac{\operatorname{Re}\lambda}{\alpha} + \mathrm{i}\,\frac{\operatorname{Im}\lambda}{\alpha\beta'} \tag{1.1}$$

inside Ω_n. The conjecture was supported by sampling single-cycle networks.

We show that this fails. Figure 1 shows the four-state counterexample Q_{w} of Section 3. Its plain uniformised eigenvalue $1 + \lambda/\alpha$ lies inside Ω_4, as it must; its rescaled eigenvalue ζ lies just outside the curved left boundary arc, while still inside the proved ellipse (disc) and polygon bounds.

1.3 The idea in one paragraph

Start from a stochastic matrix whose eigenvalue sits exactly on the boundary of Ω_n. Such matrices are known explicitly [4, 6]; some of their transitions are one-way. Turn it into a rate matrix and add weak resetting: from every state, jump with small rate proportional to ε to a random state drawn from the stationary distribution. Resetting of this kind shifts every nonzero eigenvalue by the same amount, of order ε [9]. But it also creates reverse rates of order ε on the one-way edges. A triangle with two such edges then has affinity about $2\log(1/\varepsilon)$, that is $\frac{2}{3}\log(1/\varepsilon)$ per step, so β' approaches 1 at rate $\varepsilon^{2/3}$. Dividing by β' therefore pushes the rescaled eigenvalue outward by order $\varepsilon^{2/3}$, while the eigenvalue itself has moved inward only by order ε. For small ε the outward push wins, and the rescaled point leaves Ω_n. In thermodynamic terms, the conjecture spends the affinity budget through tanh, which saturates; a weak, nearly irreversible back channel buys much affinity but little change in the dynamics.

1.4 Size of the effect, physical regime and scope

The violation is a question of shape, not a large quantitative failure: for Q_{w} the rescaled point exceeds the boundary by 0.0087 in normalised units. The driving needed is moderate. The strongest cycle of Q_{w} carries total affinity $\log(177853829/38529) \approx 8.44$, that is about $8.4\,k_{\mathrm{B}}T$, or $2.8\,k_{\mathrm{B}}T$ per step, well below the roughly $20\,k_{\mathrm{B}}T$ of one ATP hydrolysis under cellular conditions. The violating mode is poorly coherent: $|\operatorname{Im}\lambda|/|\operatorname{Re}\lambda| = 450\sqrt{3}/4745 \approx 0.16$. All constructions produce violations near the left boundary arcs of Ω_n, i.e. for strongly damped modes, so they do not touch the high-coherence regime of biochemical clocks.

The practical consequence concerns inference. If the conjectured region were used to infer a minimum affinity from the eigenvalue of Q_{w}, it would demand a per-step affinity $M \approx 3.075$, whereas the true value is $M \approx 2.812$: an overestimate of about 9% (Section 8). Table 1 summarises which bounds remain safe.

Scope matters. Every counterexample in this paper lives on the complete graph: every pair of states is connected in both directions. The "exact dimension threshold" of the title refers to the universal claim over all bidirectional networks of a given size: universal containment holds for

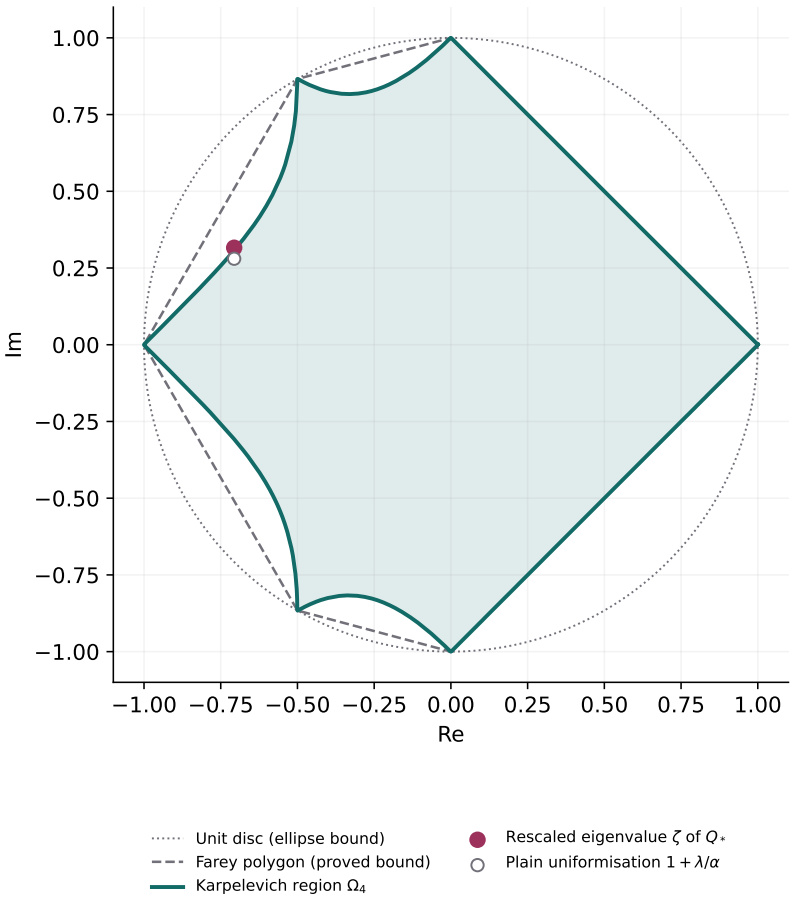


Figure 1: The Karpelevich region Ω_4 (shaded), the unit disc (dotted; in these coordinates the proved affinity ellipse of [7] becomes the unit disc), and the Farey polygon, the convex hull of the roots of unity of order at most four (dashed; the proved polygon bound of [7, Theorem 5]). The open circle is the plain uniformised eigenvalue $1 + \lambda_{\mathrm{w}}/\alpha_{\mathrm{w}}$ of the witness Q_{w}; the filled dot is its rescaled eigenvalue (1.1), which lies outside Ω_4 but inside both proved bounds. A magnified view is Figure 2.

$n \leq 3$ and fails for every $n \geq 4$. The single-cycle (ring) networks that Xu et al. used as evidence are not covered; we record them as an open problem with numerical evidence.

1.5 Roadmap and what is new

Section 2 quotes the conjecture and reduces it to a containment (C). Section 3 gives the four-state witness and its integer certificate. Section 4 proves containment for $n \leq 3$. Sections 5 and 6 explain the witness as one member of a resetting family and give an explicit interval of counterexamples. Section 7 extends the construction to every $n \geq 4$ and gives exactly certified five- and six-state instances. Section 8 discusses consequences and open problems.

The boundary-realising matrices [4, 6] and the uniform spectral shift under resetting [9] are known. The new contributions are the exact positive-rate witness, the ε versus $\varepsilon^{2/3}$ mechanism, the explicit interval, the certified instances in five and six states, the extension to every $n \geq 4$, and the proof of containment for $n \leq 3$.

Bound on the rescaled spectrum	Status	Source
Sector (coherence) bound	proved	[1] (conjectured), [8] (proof)
Affinity ellipse (unit disc in ζ)	proved	[10] (conjectured), [7]
Farey polygon	proved	[7, Theorem 5]
Affine Karpelevich region (S3)	true for $n \le 3$; false for every $n \ge 4$	this paper
Affine Karpelevich, single-cycle networks	open; no violation found numerically	Problem 8.1

Table 1: Which spectral bounds can be used for affinity inference.

2 The conjecture and the containment tested

2.1 Rate matrices and affinities

We use the row convention. A rate matrix is a real matrix $Q = (q_{ij})_{i,j=1}^{n}$ with

$$q_{ij} \ge 0 \quad (i \ne j), \qquad q_{ii} = -\sum_{j \ne i} q_{ij}. \tag{2.1}$$

For $n \ge 2$ we assume that Q is irreducible and has bidirectional support: $q_{ij} > 0$ if and only if $q_{ji} > 0$. The largest escape rate

$$\alpha(Q) = \max_i(-q_{ii}) > 0 \tag{2.2}$$

sets the time unit: $I + Q/\alpha$ is a stochastic matrix, so ordinary uniformisation places $1 + \lambda/\alpha$ in Ω_n for every $\lambda \in \operatorname{spec}(Q)$.

For an oriented simple cycle $c = (i_1, \dots, i_m, i_1)$ in the support,

$$F_c = \log \prod_{k=1}^{m} \frac{q_{i_k i_{k+1}}}{q_{i_{k+1} i_k}}, \qquad M(Q) = \max_c \frac{|F_c|}{|c|}, \qquad \beta(Q) = \tanh \frac{M(Q)}{2}, \tag{2.3}$$

with $M = 0$ if all affinities vanish. Thus M is the largest affinity per step, and $\beta \in [0, 1)$ measures how far from equilibrium the most driven cycle is. Two-cycles have zero affinity. Allowing repeated closed walks gives the same maximum: a closed walk decomposes into simple cycles, and its mean affinity cannot exceed the largest simple-cycle mean. Reversing the orientation of c changes the sign of F_c; since the maximum in [12] runs over both orientations, $\max_c F_c/|c| = \max_c |F_c|/|c|$. The source uses the column convention; transposing Q preserves the spectrum and M. If edge forces are defined with stationary fluxes $q_{ij}\pi_i/(q_{ji}\pi_j)$, the stationary factors cancel around each cycle and give the same F_c.

Our β is the parameter β' of [12, Supplement]. It differs from the parameter β of that paper's main theorem, which uses the largest edge force $\tanh(\max_e F_e/2)$.

2.2 The conjecture, verbatim

Let

$$\Omega_n = \bigcup \{\operatorname{spec}(P) : P \in \mathbb{R}^{n \times n},\ P \ge 0,\ P\mathbf{1} = \mathbf{1}\}. \tag{2.4}$$

Write R for a rate matrix in the column convention of [12], with $\alpha = \max_i |R_{ii}|$ and $\beta' = \tanh(\max_c\{\mathcal{F}_c/(2n_c)\})$, where n_c is the length of cycle c. The supplement of [12] (Section I.2) states:

> *Conjecture.* The region Ω'_n of eigenvalues of the Markov rate matrix R is the affine image of Ω_n under the transformation:
>
> $$\Omega'_n = \{\alpha(\operatorname{Re} z - 1) + \mathrm{i}\alpha\beta' \operatorname{Im} z \mid z \in \Omega_n\}. \qquad \text{(S3)}$$

Here Ω'_n is the set of eigenvalues of rate matrices of size n, and α, β' are the values computed from R. Two readings are possible: for each R, $\mathrm{spec}(R)$ lies in the right side; or the union of spectra over all rate matrices with given (α, β') equals the right side. Both readings contain the inclusion "$\subseteq$" for each individual matrix, and this is all we use. Define

$$T_{\alpha,\beta}(z) = \alpha(\mathrm{Re}\, z - 1) + \mathrm{i}\alpha\beta\, \mathrm{Im}\, z. \tag{2.5}$$

Proposition 2.1 ((S3) implies (C))**.** *Under either reading, (S3) implies that every rate matrix Q of size n satisfies*

$$\mathrm{spec}(Q) \subseteq T_{\alpha(Q),\beta(Q)}(\Omega_n). \tag{C}$$

Hence any rate matrix violating (C) *refutes (S3).*

Proof. Let $R = Q^{\mathsf{T}}$. Then $\mathrm{spec}(R) = \mathrm{spec}(Q)$, α is unchanged, and $\beta'(R) = \beta(Q)$ by the orientation remark above. Each eigenvalue of R belongs to Ω'_n under either reading, and (S3) says $\Omega'_n = T_{\alpha,\beta'}(\Omega_n)$. □

Refuting (C) therefore needs no interpretation of which boundary points are attained. When $\beta > 0$, (C) says that every eigenvalue has rescaled coordinate

$$\zeta_Q(\lambda) = 1 + \frac{\mathrm{Re}\,\lambda}{\alpha} + \mathrm{i}\frac{\mathrm{Im}\,\lambda}{\alpha\beta} \in \Omega_n. \tag{2.6}$$

In words: undo the time scale and the affinity flattening, then ask whether the result is an eigenvalue of some stochastic matrix. When $\beta = 0$, all affinities vanish, Kolmogorov's criterion gives detailed balance, the spectrum is real, and (C) holds because $[-1, 1] \subseteq \Omega_n$ for $n \geq 2$.

2.3 What is known about the boundary of Ω_n

Let $\mathcal{F}_n$ be the Farey fractions of order n, the reduced fractions $p/q \in [0, 1]$ with $q \leq n$. Two fractions are *Farey neighbours* if they are consecutive in $\mathcal{F}_n$.

Theorem 2.2 (Karpelevič; Ito's form [5, 3], [4, Theorem 2.1])**.** *Ω_n is closed, symmetric about the real axis and contained in the unit disc. It meets the unit circle exactly at the points $e^{2\pi \mathrm{i} p/q}$, $p/q \in \mathcal{F}_n$. Its boundary consists of these points and of curvilinear arcs joining them in circular order. If the endpoints of an arc are $e^{2\pi \mathrm{i} p/q}$ and $e^{2\pi \mathrm{i} r/s}$, with Farey neighbours $p/q, r/s$ and $q < s$, the arc is given by*

$$\lambda^s\big(\lambda^q - \tau\big)^{\lfloor n/q \rfloor} = (1 - \tau)^{\lfloor n/q \rfloor}\lambda^{q\lfloor n/q \rfloor}, \qquad \tau \in [0, 1]. \tag{2.7}$$

Equation (2.7) is Eq. (S2) of [12], with $(b_1, b_2) = (q, s)$; in [4] the parameter is written $\beta = 1 - \alpha$. We read "arcs joining them in circular order" in the standard way: each arc is a simple curve, and the arc point is a continuous function of τ, equal to $e^{2\pi \mathrm{i} r/s}$ at $\tau = 0$ and to $e^{2\pi \mathrm{i} p/q}$ at $\tau = 1$. Johnson and Paparella [4, Theorem 3.2] realise every arc by an explicit stochastic matrix whose characteristic polynomial is the reduced form of (2.7); Kirkland and Šmigoc [6] describe the realising matrices in detail. We need three further elementary facts.

Lemma 2.3 (Star shape)**.** *Let $n \geq 2$. If $z \in \Omega_n$ and $0 \leq r \leq 1$, then $rz \in \Omega_n$.*

Proof. Let P be stochastic with $Pu = zu$, $u \neq 0$, and let $\pi \geq 0$ be a stationary row vector, $\pi P = \pi$, $\pi\mathbf{1} = 1$ (it exists by the Perron–Frobenius theorem). If $z = 1$, the $n \times n$ stochastic matrix whose first two rows are $(1, 0, 0, \ldots, 0)$ and $(1 - r, r, 0, \ldots, 0)$ and whose remaining rows equal $(1, 0, \ldots, 0)$ is lower triangular with eigenvalue r, so $r \in \Omega_n$; this needs $n \geq 2$, and indeed $\Omega_1 = \{1\}$. If $z \neq 1$, then $\pi u = \pi P u = z\pi u$ forces $\pi u = 0$. The matrix $P_r = rP + (1 - r)\mathbf{1}\pi$ is stochastic and $P_r u = rzu$. (Irreducibility of P is not needed.) This interpolation is a standard device; we record it for completeness. □

Lemma 2.4 (Farey neighbours)**.** *If $a/b < c/d$ are reduced fractions in $[0,1]$ with $bc - ad = 1$ and $b + d > n$, then no fraction with denominator at most n lies strictly between them; so they are Farey neighbours of order n when $b, d \le n$.*

Proof. If $a/b < x/y < c/d$, then $bx - ay \ge 1$ and $cy - dx \ge 1$, so $y = y(bc - ad) = b(cy - dx) + d(bx - ay) \ge b + d > n$. □

Lemma 2.5 (Radial exterior test)**.** *Let $W \subset \mathbb{C} \setminus \{0\}$ and suppose $S \ge 1/|w|$ for all $w \in W$. If, for every pair of Farey neighbours of order n, the polynomial difference of the two sides of (2.7) has no zero $\lambda = \sigma w$ with $w \in W$, $1 \le \sigma \le S$ and $\tau \in [0,1]$, then $W \cap \Omega_n = \emptyset$.*

Proof. Suppose $w \in W \cap \Omega_n$. Since Ω_n is closed and lies in the unit disc, $\rho = \max\{r : rw/|w| \in \Omega_n\}$ exists and $|w| \le \rho \le 1$. The point $\rho w/|w|$ is a boundary point of Ω_n, because points further out on the same ray are not in Ω_n. By Theorem 2.2 it is a root of unity of order at most n or a point of an arc; in both cases it solves (2.7) for some Farey pair and some $\tau \in [0,1]$ (the endpoints correspond to $\tau \in \{0,1\}$). It equals σw with $\sigma = \rho/|w| \in [1, 1/|w|] \subseteq [1, S]$, contradicting the hypothesis. □

Lemma 2.5 uses only the description of the boundary in Theorem 2.2, not the identification of a particular root branch. It is the basis of the computer certificates in Section 7.4.

2.4 The left boundary arc in four states

Lemma 2.6 (Left boundary in dimension four)**.** *Let*

$$g(x) = 3x^2 - 2x - 1 + 4x\sqrt{-2x-1}, \qquad -1 \le x \le -\tfrac{1}{2}. \tag{2.8}$$

The upper half of the boundary arc of Ω_4 joining -1 and $e^{2\pi \mathrm{i}/3}$ is the graph $y = \sqrt{g(x)}$, $-1 \le x \le -\frac{1}{2}$. Every $x + \mathrm{i}y \in \Omega_4$ with $-1 \le x \le -\frac{1}{2}$ satisfies $y^2 \le g(x)$.

Proof. The arc. For the Farey neighbours $1/3, 1/2$ we have $q = 2$, $s = 3$, $\lfloor 4/2 \rfloor = 2$, and (2.7) becomes, after division by λ^3,

$$(z^2 - \tau)^2 = (1 - \tau)^2 z. \tag{2.9}$$

Removing the root $z = 1$ leaves the real cubic $z^3 + z^2 + (1 - 2\tau)z - \tau^2 = 0$. A real cubic has at most one root in the open upper half-plane, so for each τ every nonreal upper root of (2.9) is *the* upper root of this cubic; no branch choice is involved. (For $\tau = a^3 - a^2 + a \in (0,1)$ the cubic's only real root is $a^2 > 0$, see Section 5, so its other two roots are a nonreal conjugate pair.) Section 5 shows that with $\tau = a^3 - a^2 + a$, a strictly increasing bijection of $(0,1)$, this root is $x_0 + \mathrm{i}y_0$ with $x_0 = -(1 + a^2)/2$ and $y_0^2 = g(x_0)$ (equations (5.4)–(5.5); g arises by eliminating τ from the coefficient relations $1 - 2\tau = y^2 - 3x^2 - 2x$, $\tau^2 = (-1 - 2x)(x^2 + y^2)$, and taking the branch that vanishes at $x = -1$). As a runs over $(0,1)$, x_0 decreases from $-1/2$ to -1. So the upper arc is the graph of $\sqrt{g}$ over $[-1, -\frac{1}{2}]$.

The slice. Let $U = \{x + \mathrm{i}y : -1 < x < -\frac{1}{2},\ y > \sqrt{g(x)}\}$. It is connected. We show that no point of U solves (2.7) for any Farey pair of order 4 and any $\tau \in [0,1]$; by Theorem 2.2 every boundary point does, so $U \cap \partial\Omega_4 = \emptyset$. Since $y > 0$ on U, it is enough to look for upper roots. The pairs are $0/1, 1/4$; $1/4, 1/3$; $1/3, 1/2$ and their reflections $p/q \mapsto 1 - p/q$, which give the same equations.

- $(q, s) = (1, 4)$: for $\lambda \ne 0$ the equation is $(\lambda - \tau)^4 = (1 - \tau)^4$, so $\lambda = \tau + (1 - \tau)\omega$ with $\omega^4 = 1$. The nonreal roots have real part $\tau \ge 0$.

- $(q,s)=(3,4)$: the equation is $\lambda^3(\lambda^4-\tau\lambda-(1-\tau))=0$, and for $\lambda\neq 1$ it says $\tau=\lambda^3+\lambda^2+\lambda+1$. If $\lambda=x+\mathrm{i}y$, $y>0$, this is real only if $y^2=3x^2+2x+1$. Every root has $|\lambda|\le 1$, because $|\lambda|>1$ gives $|\lambda|^4>\tau|\lambda|+1-\tau\ge|\tau\lambda+1-\tau|$. Then $x^2+y^2=4x^2+2x+1\le 1$ forces $x\ge-\frac{1}{2}$.
- $(q,s)=(2,3)$: by the first part, the only upper root is on the graph $y=\sqrt{g(x)}$ when $\tau\in(0,1)$; $\tau=0$ gives $e^{2\pi\mathrm{i}/3}$ ($x=-\frac{1}{2}$) and $\tau=1$ gives ± 1.

None of these points lies in U. So the connected set U lies either in the interior of Ω_4 or in its complement. It contains points of modulus greater than one, so $U\cap\Omega_4=\emptyset$. At the endpoints the unit-disc containment gives the inequality directly: at $x=-1$, $y^2\le 1-1=0=g(-1)$, and at $x=-\frac{1}{2}$, $y^2\le\frac{3}{4}=g(-\frac{1}{2})$. □

3 A four-state counterexample

The witness below is 2700 times a member of the resetting family of Section 5: a boundary-realising stochastic matrix plus weak resetting at $\varepsilon=1/100$. Resetting barely moves the eigenvalue, but makes the triangle $1\to 2\to 3$ strongly (yet not maximally) driven, so the affinity flattening undershoots. Readers can verify the counterexample first and see the mechanism afterwards.

Theorem 3.1 (Exact four-state failure)**.** *The rate matrix*

$$Q_{\mathrm{w}}=\begin{pmatrix}-2768 & 2727 & 27 & 14\\ 1427 & -2768 & 1327 & 14\\ 1327 & 27 & -2768 & 1414\\ 27 & 27 & 2727 & -2781\end{pmatrix} \tag{3.1}$$

has positive off-diagonal rates and violates (C)*.*

Proof. We check the spectrum, the affinity scale, and the separation from Ω_4.

Spectrum. Every row sums to zero, $\alpha_{\mathrm{w}}=2781$, the stationary distribution is $(27,27,27,14)/95$, and direct expansion gives

$$\det(zI-Q_{\mathrm{w}})=z(z+1595)\big((z+4745)^2+607500\big). \tag{3.2}$$

Since $607500=450^2\cdot 3$,

$$\lambda_{\mathrm{w}}=-4745+450\sqrt{3}\,\mathrm{i} \tag{3.3}$$

is a simple eigenvalue.

Affinity scale. Up to reversal, the complete graph on four vertices has four simple three-cycles and three simple four-cycles. Table 2 lists their ratios $K_c=\exp|F_c|$. The largest mean is attained by the cycle 123, so exactly

$$\beta_{\mathrm{w}}=\frac{K^{1/3}-1}{K^{1/3}+1},\qquad K=\frac{177853829}{38529}. \tag{3.4}$$

Each ratio is below $19^{|c|}$, checked by the integer column of the table. Therefore

$$M(Q_{\mathrm{w}})<\log 19,\qquad \beta_{\mathrm{w}}<\tanh\frac{\log 19}{2}=\frac{9}{10}. \tag{3.5}$$

Separation. Write $\zeta_{Q_{\mathrm{w}}}(\lambda_{\mathrm{w}})=x+\mathrm{i}y$. By (2.6), (3.3) and (3.5),

$$x=-\frac{1964}{2781},\qquad y^2>\frac{607500}{2781^2}\left(\frac{10}{9}\right)^2=\frac{250000}{2577987}=\frac{750000}{7733961}=:Y. \tag{3.6}$$

Cycle	Length	K_c	$19^{\lvert c\rvert}\,\mathrm{den}(K_c) - \mathrm{num}(K_c)$
123	3	177853829/38529	86416582
124	3	2727/1427	9785066
134	3	1327/27	183866
234	3	1327/27	183866
1234	4	134027/1427	185834040
1243	4	134027/1427	185834040
1324	4	1760929/729	93243080

Table 2: All nontrivial simple-cycle affinity ratios of Q_{w}, in lowest terms. Cycle labels include the closing edge.

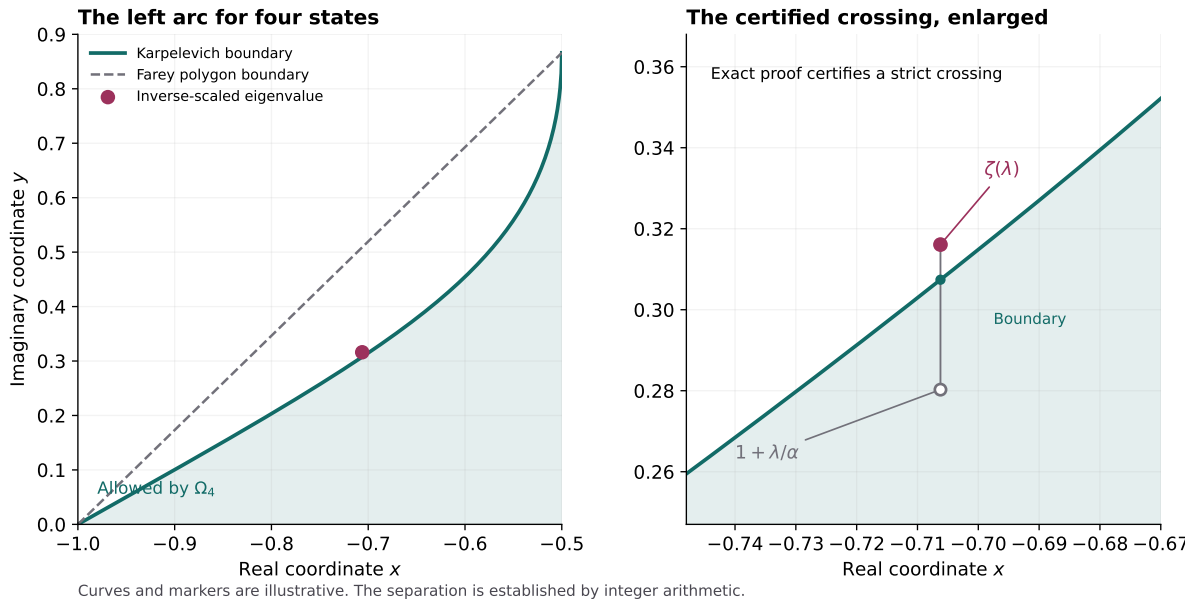


Figure 2: Left: the left boundary arc of Ω_4 (solid) and the edge of the Farey polygon (dashed). The rescaled eigenvalue of Q_{w} lies outside the arc. Right: magnified view, showing the plain uniformised point $1 + \lambda_{\mathrm{w}}/\alpha_{\mathrm{w}}$ inside and the rescaled point outside. The separation is proved by the integer inequality (3.8); the figure is illustrative.

Here $-1 < x < -1/2$, and (2.8) gives

$$g(x) = \frac{14761695 - 7856\sqrt{3189807}}{7733961}. \tag{3.7}$$

Thus $g(x) < Y$ is equivalent to $7856\sqrt{3189807} > 14011695$; both sides are positive, and

$$7856^2 \cdot 3189807 - 14011695^2 = 536879736927 > 0. \tag{3.8}$$

Hence $y^2 > Y > g(x)$, and $x + \mathrm{i}y \notin \Omega_4$ by Lemma 2.6. □

No numerical approximation enters the proof. For scale, $\beta_{\mathrm{w}} \approx 0.8866882$, $x \approx -0.7062208$, $y \approx 0.3160830$ and $\sqrt{g(x)} \approx 0.3074076$: the point exceeds the boundary height by about 0.0087. Its modulus, about 0.774, is less than one, as the proved ellipse bound requires. The certificate script of Section 7.4 also confirms the separation through the radial test of Lemma 2.5, independently of Lemma 2.6.

4 Three states are not enough

Theorem 4.1 (Containment in dimensions at most three)**.** *Every irreducible bidirectional rate matrix of size $n \le 3$ satisfies* (C)*. Hence, within this class, four is the smallest dimension in which the containment can fail.*

Proof. For $n = 1$, $Q = 0$. For $n = 2$ the spectrum is real, and uniformisation places $1 + \lambda/\alpha$ in $[-1, 1] = \Omega_2 \cap \mathbb{R}$; the factor β does not move real points.

Let $n = 3$. Real eigenvalues are handled as before. A nonreal pair $a \pm \mathrm{i}b$ requires $\beta > 0$ (otherwise the spectrum is real), and the third eigenvalue is zero. Since all escape rates are at most α, $2a = \operatorname{tr} Q \ge -3\alpha$, so $x = 1 + a/\alpha \ge -\frac{1}{2}$. The sector bound conjectured by Barato and Seifert [1] and proved by Ohga, Ito and Kolchinsky [8, Eq. (10)] states, for every nonreal eigenvalue of a rate matrix,

$$\frac{|b|}{-a} \le \max_c \tanh \frac{|F_c|}{2|c|} \cot \frac{\pi}{|c|}, \tag{4.1}$$

the maximum running over simple cycles with the affinity of (2.3); the bound is invariant under transposition, so the row/column convention is immaterial. Two-cycles contribute $\tanh(0) \cot(\pi/2) = 0$. In three states every other simple cycle has length three, so the right side equals $\beta \cot(\pi/3) = \beta/\sqrt{3}$, and the rescaled coordinate obeys

$$|y| = \frac{|b|}{\alpha\beta} \le \frac{-a}{\alpha\sqrt{3}} = \frac{1 - x}{\sqrt{3}}, \qquad -\frac{1}{2} \le x \le 1. \tag{4.2}$$

This is the triangle with vertices $1, e^{\pm 2\pi\mathrm{i}/3}$. Every point of it is an eigenvalue of $p_0 I + p_1 C + p_2 C^2$, where C is the cyclic permutation matrix and (p_0, p_1, p_2) is a probability vector. Hence the triangle lies in Ω_3. □

Remark 4.2. This concerns containment only. It does not assert that every point of the affine image is attained for given (α, β).

5 Where the witness comes from: weak resetting

5.1 A boundary matrix

Choose $a \in (0, 1)$, put

$$t = a^3 - a^2 + a \in (0, 1), \tag{5.1}$$

and define

$$P_0(t) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ t & 0 & 1 - t & 0 \\ 1 - t & 0 & 0 & t \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{5.2}$$

This is the Johnson–Paparella Type II realising matrix $\alpha X + \beta Y$ for the Farey neighbours $1/3, 1/2$ with $\beta = t$, $\alpha = 1 - t$ [4, Theorem 3.2], after the relabelling of states $(1, 2, 3, 4) \mapsto (3, 4, 2, 1)$. Its characteristic polynomial is the boundary polynomial

$$\det(zI - P_0) = (z^2 - t)^2 - (1 - t)^2 z, \tag{5.3}$$

one root of the cubic factor is a^2, and

$$\operatorname{spec}(P_0) = \{1, a^2, x_0 \pm \mathrm{i}y_0\}, \tag{5.4}$$

$$x_0 = -\frac{1 + a^2}{2}, \qquad y_0 = \frac{1 - a}{2}\sqrt{3a^2 - 2a + 3} > 0, \qquad y_0^2 = g(x_0). \tag{5.5}$$

These follow from the sum and product of the roots.

5.2 Add resetting that preserves the stationary distribution

Let $v = (1, 1, 1, t)$. Then $vP_0 = v$ and $v\mathbf{1} = 3 + t$. For $\varepsilon > 0$ set

$$R_\varepsilon = P_0 - I + \varepsilon\{\mathbf{1}v - (3+t)I\}. \tag{5.6}$$

The added term is resetting at rate $(3+t)\varepsilon$ to the stationary distribution $v/(3+t)$. Biochemically, resetting to the stationary mix is the same as turnover or dilution: a molecule is degraded and replaced by a fresh one drawn from the steady-state population. The off-diagonal rates of R_ε are

$$\begin{pmatrix} \cdot & 1+\varepsilon & \varepsilon & t\varepsilon \\ t+\varepsilon & \cdot & 1-t+\varepsilon & t\varepsilon \\ 1-t+\varepsilon & \varepsilon & \cdot & t(1+\varepsilon) \\ \varepsilon & \varepsilon & 1+\varepsilon & \cdot \end{pmatrix}, \tag{5.7}$$

all positive; the escape rates are $1 + (2+t)\varepsilon$ (three times) and $1 + 3\varepsilon$, so $\alpha_\varepsilon = 1 + 3\varepsilon$.

If $P_0 u = zu$ with $z \neq 1$, then $(z-1)vu = 0$, so $vu = 0$ and $R_\varepsilon u = (z - 1 - (3+t)\varepsilon)u$. Hence

$$\lambda_\varepsilon = x_0 - 1 - (3+t)\varepsilon + \mathrm{i}y_0 \tag{5.8}$$

is an eigenvalue: the uniform shift of [9]. At $a = 2/3$, $t = 14/27$, $\varepsilon = 1/100$ one has exactly $Q_{\mathrm{w}} = 2700 R_{1/100}$, which explains (3.2).

5.3 The dominant cycle

Put

$$A_\varepsilon = \frac{1+\varepsilon}{t+\varepsilon} > 1, \qquad B_\varepsilon = \frac{1-t+\varepsilon}{\varepsilon} > 1. \tag{5.9}$$

The seven cycle ratios take the forms of Table 3.

Cycle(s)	Length	Larger forward/backward ratio
123	3	$A_\varepsilon B_\varepsilon^2$
124	3	A_ε
134, 234	3	B_ε
1234, 1243	4	$A_\varepsilon B_\varepsilon$
1324	4	B_ε^2

Table 3: Cycle ratios throughout the resetting family.

Since $\log A_\varepsilon, \log B_\varepsilon > 0$, the mean $(\log A_\varepsilon + 2\log B_\varepsilon)/3$ exceeds every other mean; for example its excess over the four-cycle means is $(\log A_\varepsilon + 5\log B_\varepsilon)/12$ and $(2\log A_\varepsilon + \log B_\varepsilon)/6$. So 123 dominates for every $\varepsilon > 0$, and with $w_\varepsilon = (A_\varepsilon B_\varepsilon^2)^{-1/3}$,

$$\beta_\varepsilon = \frac{1 - w_\varepsilon}{1 + w_\varepsilon}, \qquad x_\varepsilon = \frac{x_0 - t\varepsilon}{1 + 3\varepsilon}, \qquad y_\varepsilon = \frac{y_0}{(1+3\varepsilon)\beta_\varepsilon}. \tag{5.10}$$

5.4 Two perturbation orders

Theorem 5.1 (Counterexamples near every point of the open arc)**.** *Fix $a \in (0,1)$ and t by* (5.1)*. Then*

$$y_\varepsilon^2 - g(x_\varepsilon) = 4y_0^2 \left(\frac{t}{(1-t)^2}\right)^{1/3} \varepsilon^{2/3} + O_a(\varepsilon) \quad (\varepsilon \downarrow 0), \tag{5.11}$$

where the implied constant depends on a. Hence there is $\varepsilon_0(a) > 0$ such that R_ε violates (C) *for all $0 < \varepsilon < \varepsilon_0(a)$.*

Proof. From (5.9), $w_\varepsilon = (t/(1-t)^2)^{1/3}\varepsilon^{2/3}(1+O(\varepsilon))$, so $\beta_\varepsilon^{-2} = 1+4(t/(1-t)^2)^{1/3}\varepsilon^{2/3}+O(\varepsilon^{4/3})$. The factor $(1+3\varepsilon)^{-2}$ changes by $O(\varepsilon)$, $x_\varepsilon = x_0+O(\varepsilon)$, and g is smooth near the interior point x_0 with $g(x_0) = y_0^2$. Subtraction gives (5.11), whose leading coefficient is positive; since $\varepsilon = o(\varepsilon^{2/3})$, the gap is positive for small ε, and Lemma 2.6 applies. □

Two edges of the dominant triangle have no reverse transition at $\varepsilon = 0$. Reverse rates of order ε make its ratio grow like ε^{-2}; averaging over three edges gives the exponent 2/3. The eigenvalue moves linearly in ε. As $\varepsilon \to 0$, $\beta_\varepsilon \to 1$ and the rescaled point returns to the boundary point $x_0 + \mathrm{i}y_0$, so the limit itself is not a counterexample.

6 An explicit interval of counterexamples

Theorem 5.1 gives a possibly different neighbourhood of $\varepsilon = 0$ for each point of the arc. For the witness's parameter we give one explicit interval.

Theorem 6.1 (Uniform interval of counterexamples)**.** *Let $a = 2/3$, $t = 14/27$. For every $0 < \varepsilon \le 1/100$,*

$$y_\varepsilon^2 - g(x_\varepsilon) \ge \frac{16505}{1527696}\varepsilon^{2/3} > 0, \tag{6.1}$$

so every R_ε in this interval violates (C)*.*

Proof. Here $x_0 = -13/18$ and $y_0^2 = 1/12$. First, $A_\varepsilon \le 1/t$ and $1 - t + \varepsilon \le 1327/2700$, so

$$(A_\varepsilon B_\varepsilon^2)\varepsilon^2 \le \frac{1}{t}\left(\frac{1327}{2700}\right)^2 = \frac{1760929}{3780000} < \frac{64}{125},$$

hence $w_\varepsilon \ge (5/4)\varepsilon^{2/3}$. Since $0 < w_\varepsilon < 1$, $\beta_\varepsilon^{-2} = ((1+w_\varepsilon)/(1-w_\varepsilon))^2 \ge 1 + 4w_\varepsilon$. Using $(1+3\varepsilon)^{-2} \ge 1 - 6\varepsilon$ for the baseline term and $(1+3\varepsilon)^{-2} \ge 10000/10609$ for the correction,

$$y_\varepsilon^2 \ge \frac{1}{12} - \frac{\varepsilon}{2} + \frac{12500}{31827}\varepsilon^{2/3}. \tag{6.2}$$

Next, $0 \le x_\varepsilon - x_0 = \frac{89}{54}\frac{\varepsilon}{1+3\varepsilon} \le \frac{89}{54}\varepsilon$ and $x_\varepsilon \in [-13/18, -1964/2781]$. With $s = \sqrt{-2x-1}$ one has $16/25 < s \le 2/3$, because $\frac{1147}{2781} - (\frac{16}{25})^2 = \frac{4939}{1738125} > 0$, and $g'(x) = -3s^2+6s-5+2/s$. Its derivative in s is at most $54/25 - 9/2 < 0$ on this interval, so $g'(x) \le -3(16/25)^2+6(16/25)-5+2/(16/25) = 3681/5000 < 3/4$, and the mean value theorem gives

$$g(x_\varepsilon) \le \frac{1}{12} + \frac{89}{72}\varepsilon. \tag{6.3}$$

Therefore $y_\varepsilon^2 - g(x_\varepsilon) \ge \frac{12500}{31827}\varepsilon^{2/3} - \frac{125}{72}\varepsilon$. Finally $\varepsilon^{1/3} < 11/50$ because $(11/50)^3 = 1331/125000 > 1/100$, and $\frac{12500}{31827} - \frac{125}{72}\cdot\frac{11}{50} = \frac{16505}{1527696}$. □

The constant in (6.1) is conservative; a dense sweep gives a minimum of $(y_\varepsilon^2 - g(x_\varepsilon))/\varepsilon^{2/3}$ near 0.117.

Corollary 6.2 (An open set of positive-rate counterexamples)**.** *There is an open neighbourhood of Q_{w} in $\mathbb{R}^{12}$, the space of off-diagonal rates, consisting of rate matrices that violate* (C)*.*

Proof. The rates of Q_{w} are positive, its complex pair is simple, $\beta_{\mathrm{w}} > 0$, and $-1 < x < -1/2$. The eigenvalues near this simple pair, the maximal escape rate, and the maximum of the finitely many cycle means depend continuously on the rates. Since Ω_4 is closed and the transformed point lies outside it, it stays outside nearby. □

7 Five, six and every larger number of states

Appending states to a four-state example does not suffice, because Ω_n grows with n. We repeat the construction with the boundary matrices of each dimension. Throughout, the order of limits is: fix n; then fix $t \in (0, \eta_n)$; then let $\varepsilon \to 0$. No threshold is uniform in t or in n.

7.1 Boundary matrices in every dimension

For even $n = 2m \geq 4$ let the nonzero entries of $P_n(t)$ be

$$P_{2k-1,2k} = 1,\ P_{2k,2k-1} = t\ (1 \leq k \leq m), \qquad P_{2k,2k+1} = 1 - t\ (1 \leq k < m), \qquad P_{n,2} = 1 - t. \tag{7.1}$$

For odd $n = 2m + 1 \geq 5$ let them be

$$P_{1,2} = 1,\ P_{2k,2k+1} = 1,\ P_{2k+1,2k} = t\ (1 \leq k \leq m),\ P_{2k+1,2k+2} = 1-t\ (1 \leq k < m),\ P_{n,1} = 1-t. \tag{7.2}$$

These are irreducible stochastic matrices for $0 < t < 1$, and direct expansion gives

$$\det(zI - P_n(t)) = F_{n,t}(z) := \begin{cases} (z^2 - t)^m - (1 - t)^m z, & n = 2m, \\ z(z^2 - t)^m - (1 - t)^m, & n = 2m + 1, \end{cases} \tag{7.3}$$

with stationary row vector $v = (c, 1, \ldots, 1)$, $V = v\mathbf{1} = n - 1 + c$, where $c = t$ for even and $c = 1 - t$ for odd n.

Lemma 7.1 (Which arc)**.** *Let $s = n-1$ for even n and $s = n$ for odd n, and $r = (s-1)/2$. Then r/s and $1/2$ are Farey neighbours of order n, and (2.7) for this pair, with $\tau = t$, is $F_{n,t}(\lambda) = 0$ up to a power of λ. The pair is of Johnson–Paparella Type II ($s < 2\lfloor n/2 \rfloor$) for even n and Type III ($s > 2\lfloor n/2 \rfloor$, $d = s - 2\lfloor n/2 \rfloor = 1$) for odd n [4, proof of Theorem 3.2], and the characteristic polynomial of $P_n(t)$ is the reduced Type II/III boundary polynomial. (We do not claim, and nothing below uses, that $P_n(t)$ is the Johnson–Paparella realising matrix itself up to a relabelling of states.)*

Proof. s is odd and $s \cdot 1 - 2r = 1$; since $s + 2 > n$, Lemma 2.4 applies. In (2.7) take $q = 2$ (the smaller denominator) and $\lfloor n/2 \rfloor = m$. For $n = 2m$, $s = 2m - 1$: $\lambda^{2m-1}(\lambda^2 - t)^m = (1 - t)^m \lambda^{2m}$, i.e. $\lambda^{2m-1}[(\lambda^2 - t)^m - (1 - t)^m \lambda] = 0$. For $n = 2m+1$, $s = 2m+1$: $\lambda^{2m+1}(\lambda^2 - t)^m = (1 - t)^m \lambda^{2m}$, i.e. $\lambda^{2m}[\lambda(\lambda^2 - t)^m - (1 - t)^m] = 0$. The case distinction is that of [4], where Type II reads $(\lambda^q - \beta)^{\lfloor n/q \rfloor} = \alpha^{\lfloor n/q \rfloor} \lambda^{q\lfloor n/q \rfloor - s}$ and Type III reads $\lambda^d(\lambda^q - \beta)^{\lfloor n/q \rfloor} = \alpha^{\lfloor n/q \rfloor}$, with $\beta = t$, $\alpha = 1 - t$. □

Lemma 7.2 (A smooth boundary point with exterior above)**.** *Let $n \geq 4$, $\theta = \pi - \pi/s = 2\pi r/s$, $z_\circ = e^{\mathrm{i}\theta}$. There is $\eta_n > 0$ such that for every $t \in (0, \eta_n)$:*

(i) the analytic root branch $z(t) = x(t) + \mathrm{i}y(t)$ of $F_{n,t}$ with $z(0) = z_\circ$ is a simple eigenvalue of $P_n(t)$ and is the point of the boundary arc of Ω_n from $z_\circ$ to -1 at parameter $\tau = t$;

(ii) near $z(t)$, that arc is a smooth graph $y = h(x)$, and the points just above the graph lie outside Ω_n.

Proof. (i) Branch. At $t = 0$, $F_{n,0}$ is $z^{2m} - z = z(z^s - 1)$ for even n and $z^{2m+1} - 1 = z^s - 1$ for odd n; $z_\circ$ is a simple root because $z_\circ^s = 1$ and the derivative there is s, respectively $s/z_\circ$. By the implicit function theorem there are $\delta, \eta > 0$ such that, for $0 \leq t < \eta$, $F_{n,t}$ has exactly one root in the disc $|z - z_\circ| < \delta$, namely $z(t)$. Let $\kappa(\tau)$ be the arc point of Theorem 2.2 for this pair (Lemma 7.1); it is a root of $F_{n,\tau}$, depends continuously on τ and satisfies $\kappa(0) = z_\circ$. Shrinking η, $|\kappa(t) - z_\circ| < \delta$ for $t < \eta$, so $\kappa(t) = z(t)$. Being a root of the characteristic polynomial, $z(t)$ is an eigenvalue of $P_n(t)$, and it is simple for small t by continuity.

Local shape. Implicit differentiation gives

$$z'(0) = -\frac{2im}{s}\sin\theta, \qquad x''(0) = \begin{cases} \dfrac{4m(m+1)}{s^2}\cos\theta\sin^2\theta, & n = 2m, \\ \dfrac{4m(m-1)}{s^2}\cos\theta\sin^2\theta, & n = 2m+1. \end{cases} \tag{7.4}$$

Since $z'(0)$ is purely imaginary, $x'(0) = 0$; since $\cos\theta < 0 < \sin\theta$ and $m \geq 2$, $x''(0) < 0$ and $y'(0) < 0$. Hence for small $t > 0$, $x'(t) < 0$, $y'(t) < 0$, $y(t) > 0$, and the arc near $z(t)$ is a smooth graph $y = h(x)$ with $h' = y'/x'$. Moreover $\frac{d}{dt}\arg z(t)|_{t=0} = -\frac{2m}{s}\cos\theta\sin\theta > 0$, so for small t

$$x(t)y'(t) - y(t)x'(t) = |z(t)|^2\,\tfrac{d}{dt}\arg z(t) > 0. \tag{7.5}$$

(ii) Exterior side. Take a small open disc D around $z(t)$ in which the arc is the graph $y = h(x)$, and which meets no other arc: other arcs lie in other angular sectors, and the conjugate arc lies in $y < 0$, while $z(t)$ is an interior point of the arc with $\theta < \arg z(t) < \pi$. Then $D \setminus \{y = h(x)\}$ consists of two connected sets D_+ (above) and D_- (below), neither of which meets $\partial\Omega_n$; each therefore lies in the interior of Ω_n or in its complement. For small $\delta' > 0$, the point $(1-\delta')z(t)$ belongs to Ω_n by Lemma 2.3, and it lies in D_-: to first order, $(1-\delta')y - h((1-\delta')x) \approx -\delta'(y - h'x) = -\delta'(yx' - xy')/x' < 0$ by (7.5) and $x' < 0$. So $D_- \subseteq \Omega_n$. If also $D_+ \subseteq \Omega_n$, then $D \subseteq \Omega_n$ because Ω_n is closed, and $z(t)$ would be an interior point, contradicting $z(t) \in \partial\Omega_n$. Hence $D_+ \cap \Omega_n = \emptyset$. □

7.2 The leading cycle after completing the support

Fix n and t as in Lemma 7.2, and let

$$Q_{n,t,\varepsilon} = P_n(t) - I + \varepsilon(\mathbf{1}v - VI), \qquad \varepsilon > 0. \tag{7.6}$$

All off-diagonal rates are positive, $\lambda_\varepsilon = z(t) - 1 - V\varepsilon$ is an eigenvalue, and since $\min_i v_i = c$,

$$\alpha_\varepsilon = 1 + (n-1)\varepsilon, \qquad x_\varepsilon = \frac{x(t) - c\varepsilon}{1 + (n-1)\varepsilon}, \qquad y_\varepsilon = \frac{y(t)}{[1 + (n-1)\varepsilon]\beta_\varepsilon}. \tag{7.7}$$

Every simple cycle of the complete graph must be compared. Define $d_{ij} = \mathbf{1}_{\{P_{ij}>0\}} - \mathbf{1}_{\{P_{ji}>0\}} \in \{-1, 0, 1\}$. At fixed t,

$$\log\frac{q_{ij}}{q_{ji}} = d_{ij}\log(1/\varepsilon) + b_{ij} + O(\varepsilon), \tag{7.8}$$

with finite constants b_{ij} depending on t.

Lemma 7.3 (Unique critical triangle)**.** *For each $P_n(t)$, $\max_c |c|^{-1}\left|\sum_{(i,j)\in c} d_{ij}\right| = 2/3$, attained by exactly one unoriented cycle: $(n, 2, 3)$ for even n and $(n, 1, 2)$ for odd n.*

Proof. The edges with $d_{ij} = 1$ form isolated directed edges and one path of length two, $n \to 2 \to 3$ (even) or $n \to 1 \to 2$ (odd). In an oriented simple cycle with p edges of value $+1$ and q others, $q \geq 1$, and the $+1$ edges form runs of length at most two, so $p \leq 2q$ and the mean is at most $p/(p+q) \leq 2/3$; the reverse orientation is treated in the same way. Equality needs no -1 edges and all runs of length two; the unique path can be used once, so $p = 2$, $q = 1$, giving the stated triangle, whose closing edge has value zero. □

For all sufficiently small ε (depending on n and t) the critical triangle therefore has the largest mean affinity, with exact ratio

$$K_\varepsilon = \begin{cases} \dfrac{(1-t+\varepsilon)^2(1+\varepsilon)}{\varepsilon^2(t+\varepsilon)}, & n = 4, \\ \dfrac{(1-t+\varepsilon)^2}{\varepsilon^2}, & n \geq 6 \text{ even}, \\ \dfrac{(1+\varepsilon)^2}{\varepsilon^2}, & n \geq 5 \text{ odd}, \end{cases} \qquad w_\varepsilon := K_\varepsilon^{-1/3} = \kappa_n(t)\varepsilon^{2/3}(1 + O(\varepsilon)), \tag{7.9}$$

where $\kappa_4 = [t/(1-t)^2]^{1/3}$, $\kappa_n = (1-t)^{-2/3}$ for even $n \geq 6$ and $\kappa_n = 1$ for odd $n \geq 5$. For $n = 4$, $P_4(t)$ is $P_0(t)$ of (5.2) after relabelling, and κ_4 agrees with Theorem 5.1.

Theorem 7.4 (Failure in every dimension at least four)**.** *For every $n \geq 4$ there is $\eta_n > 0$ such that for each fixed $t \in (0, \eta_n)$ there is $\varepsilon_0(n,t) > 0$ with the following property: for $0 < \varepsilon < \varepsilon_0(n,t)$, $Q_{n,t,\varepsilon}$ violates* (C)*, and so does every rate matrix in an open neighbourhood of it in $\mathbb{R}^{n(n-1)}$. Together with Theorem 4.1, universal containment over irreducible bidirectional rate matrices holds exactly in dimensions $n \leq 3$.*

Proof. By (7.9), $\beta_\varepsilon^{-1} = (1+w_\varepsilon)/(1-w_\varepsilon) = 1 + 2\kappa_n \varepsilon^{2/3} + O(\varepsilon^{4/3})$, so (7.7) gives $x_\varepsilon = x(t) + O(\varepsilon)$ and $y_\varepsilon = y(t) + 2y(t)\kappa_n \varepsilon^{2/3} + O(\varepsilon)$. Since h is smooth near $x(t)$, $h(x_\varepsilon) = y(t) + O(\varepsilon)$, and

$$y_\varepsilon - h(x_\varepsilon) = 2y(t)\kappa_n(t)\varepsilon^{2/3} + O_{n,t}(\varepsilon) > 0 \tag{7.10}$$

for small ε. By Lemma 7.2(ii) the rescaled eigenvalue lies outside Ω_n. The eigenvalue is simple, the rates are positive and the separation strict, so the continuity argument of Corollary 6.2 gives an open set. □

For $n = 3$ the odd formula in (7.4) has $m = 1$ and gives $x''(0) = 0$: the pair stays on the line $x = -1/2$, the upward-exterior geometry is absent, in agreement with Theorem 4.1.

7.3 How small must ε be?

Remark 7.5 (The ε-scale)*.* The remainder in (7.10) hides constants that depend strongly on t, for two reasons. First, the constants b_{ij} in (7.8) contain $\log(1/t)$, so a long cycle with smaller valuation mean can beat the critical triangle until ε is very small. In five states the competitor is the Hamiltonian cycle $1 \to 2 \to 3 \to 4 \to 5 \to 1$, whose ratio behaves like $\varepsilon^{-3}t^{-2}$: its mean affinity $\frac{3}{5}\log\frac{1}{\varepsilon} + \frac{2}{5}\log\frac{1}{t}$ falls below the triangle's $\frac{2}{3}\log\frac{1}{\varepsilon}$ only when, up to bounded factors, $\varepsilon \lesssim t^6$ (more precisely $\varepsilon \lesssim t^6/(1-t)^3$). Second, near the arc endpoint the local boundary slope is of order $1/t$, so the $O(\varepsilon)$ term carries a factor of order $1/t$, and positivity of (7.10) needs roughly $\varepsilon^{1/3} \ll t$. For the proof's asymptotics the first condition dominates. Numerically, at $t = 1/20$ the five-, six- and seven-state gaps are still negative at $\varepsilon = 10^{-6}$. In five states the triangle becomes dominant near $\varepsilon \approx e^{-18} \approx t^6$, and the radial margin is already positive from about $\varepsilon \approx e^{-16.3}$; $\varepsilon \lesssim t^6$ is what the proof's asymptotics need, not a threshold for violation. The small-t hypothesis is a convenience of the proof: at moderate t (e.g. $t \approx 0.3$–0.6) violations appear already for $\varepsilon \approx 10^{-5}$–$10^{-4}$, which is how the certified instances below were chosen. Violations in dimension $n \geq 5$ are real but thin: margins are of order 10^{-4} or smaller.

7.4 Exactly certified counterexamples in five and six states

Theorem 7.6 (Certified instances)**.** *The rate matrices $Q_{5,3/5,1/40000}$ and $Q_{6,1/2,1/8000}$ of* (7.6) *have rational entries and positive off-diagonal rates, and both violate* (C)*.*

Proof (computer-assisted, exact). The script `certify_higher_dimensions.py` uses only integers, rational numbers and outward-rounded rational interval arithmetic (fixed-point grid 2^{-128}); no floating-point value enters any decision. For each instance it verifies:

1. row sums, positivity, $\alpha = 1+(n-1)\varepsilon$, $vP_n = v$, $\det(zI - P_n(t)) = F_{n,t}(z)$ and $\det(zI - Q) = z\,G(z + 1 + V\varepsilon)$ with $G = F_{n,t}/(z-1)$, all as exact polynomial identities;

2. a rational disc of radius below 10^{-27} containing a root z^* of $F_{n,t}$ with $\operatorname{Im} z^* > 0$, $z^* \neq 1$, by the classical bound that some root of a polynomial p of degree d lies within $d|p(z_0)|/|p'(z_0)|$ of z_0 (evaluated exactly); thus $\lambda = z^* - 1 - V\varepsilon$ is an eigenvalue of Q;

3. the exact ratio of every simple cycle of the complete graph (37 unoriented cycles for $n = 5$, 197 for $n = 6$), the dominant cycle by exact comparison $K_c^{|d|}$ versus $K_d^{|c|}$, and rational bounds $\beta_{\mathrm{lo}} \le \beta \le \beta_{\mathrm{hi}}$;

4. a rational box W of width below 10^{-37} containing the rescaled eigenvalue;

5. a rational S with $S^2 \min_{w \in W} |w|^2 \ge 1$, and the hypothesis of Lemma 2.5 for W, for all Farey pairs of order n: $[1, S] \times [0, 1]$ is bisected into closed boxes, and on each terminal box the interval extension of $(\sigma w)^s\big((\sigma w)^q - \tau\big)^k - (1-\tau)^k(\sigma w)^{qk}$, $k = \lfloor n/q \rfloor$, obtained by evaluating this expression in rectangular complex interval arithmetic with powers by binary (square-and-multiply) powering, has a real or an imaginary part that excludes 0.

Lemma 2.5 then gives $W \cap \Omega_n = \emptyset$. The terminal boxes are written to `exterior_certificates.json`. A separate checker, `check_exterior_certificate.py`, shares no code with the certifier and uses exact rational intervals without rounding. It recomputes items 1–5 from Q and the certificate: the characteristic polynomial of Q itself, the cycle ratios, the root bound, the inclusion of the image of the root disc in W, the bound on S, and, for every Farey pair, that the boxes are inside $[1, S] \times [0, 1]$, pairwise interior-disjoint and of total area $S - 1$ (so, being closed, they cover the rectangle), and that each box excludes 0. Appendix A prints the data. □

Table 4 summarises the receipts. As a control, the same exterior test certifies the four-state witness Q_{w} independently of Lemma 2.6, and fails, as it should, for points 10^{-6} below the left arc of Ω_4.

n	t	ε	dominant cycle	rescaled eigenvalue ζ	radial margin
4	14/27	1/100	$(1, 2, 3)$	$-0.706221 + 0.316083\,\mathrm{i}$	$\approx 8.7 \times 10^{-3\mathrm{a}}$
5	3/5	1/40000	$(1, 2, 5)$	$-0.849362 + 0.248186\,\mathrm{i}$	$\approx 4.7 \times 10^{-5}$
6	1/2	1/8000	$(2, 3, 6)$	$-0.874426 + 0.253551\,\mathrm{i}$	$\approx 2.2 \times 10^{-4}$

Table 4: Certified counterexamples. The radial margin is the difference between $|\zeta|$ and the modulus of the boundary point on the same ray, computed numerically for scale only ([a]for $n = 4$ the vertical excess over the arc is shown). In five and six states the runner-up cycle is a Hamiltonian cycle (respectively the five-cycle $2 \to 3 \to 4 \to 5 \to 6$), in line with Remark 7.5.

8 Consequences and open problems

8.1 The valid bounds and the size of the error

Because Ω_n lies in the unit disc, the false containment (C) would imply the affinity ellipse, but not conversely; Q_{w} separates the two. The polygon bound of [7, Theorem 5] places the rescaled spectrum in the convex hull of the roots of unity of order at most n; this polygon contains Ω_n, and Kolchinsky et al. note that it is weaker than the Karpelevich region in the irreversible limit. Our example is consistent with it, and checkable without decimals: the ratio of cycle 123 exceeds 9^3, so $\beta_{\mathrm{w}} > 4/5$, and

$$1 - x^2 - \frac{607500}{2781^2 (4/5)^2} = \frac{11709785}{30935844} > 0$$

gives $x^2 + y^2 < 1$; at this x the polygon edge is $y = (1 + x)\sqrt{3}$, and $450/\beta_{\mathrm{w}} < 1125/2 < 817$ with $1 + x = 817/2781$ puts the point below it. The refinements of [13] concern unicyclic networks and do not apply to our construction.

If (C) were used to infer an affinity from λ_{w}, it would demand

$$\beta \ge \frac{450\sqrt{3}/2781}{\sqrt{g(-1964/2781)}} \approx 0.9117,$$

that is $M = 2\operatorname{arctanh}\beta \approx 3.075$, whereas $M(Q_{\mathrm{w}}) = \frac{1}{3}\log(177853829/38529) \approx 2.812$: an overestimate of about 9.3%. The error is bounded by the gap between Ω_n and the proved polygon, so it is modest, but it is in the unsafe direction. We know of no empirical study that has used the conjectured region in this way.

8.2 Relation to other spectral regions for generators

Chruściński, Denisov, Tarnowski and Życzkowski [2] propose a rescaling of classical generators under which the eigenvalues lie in modified Karpelevič regions. As described there, that rescaling involves no cycle-affinity parameter; our results concern only the affinity flattening by β', and neither statement implies the other. Verbeken and Ginis [11] determine the spectral region of stochastic matrices supported on a directed n-cycle with self-loops, which arise from uniformised unidirectional reaction cycles; this is the irreversible single-cycle analogue of the open problem below.

8.3 Open problems

Open problem 8.1 (Single-cycle networks)**.** Does (C) hold for every rate matrix whose support is a single bidirectional cycle?

This is the setting in which (S3) was tested [12], and our mechanism does not apply to it: a ring has only one cycle, so weak reverse rates on some edges cannot create a separate, more strongly driven triangle. For uniform rings the rescaled eigenvalues are exactly the n-th roots of unity, on $\partial\Omega_n$. A multistart local search over all forward and backward rates of rings with four and five states (400 starts each, measuring the largest radial excess of any rescaled eigenvalue over a tabulated boundary of Ω_n) found no violation; the largest values, -7×10^{-5} and -3×10^{-4}, are at the resolution of the tabulation and occur near uniform rings. This is evidence, not proof, and we make no claim either way.

Open problem 8.2 (The true region)**.** Determine the exact eigenvalue region for given n, α and β, or at least the largest radial excess over Ω_n as a function of β. Is the affine containment asymptotically correct near equilibrium ($\beta \to 0$)? Does a different affinity statistic, for example one weighted by stationary flux, restore a Karpelevich-shaped bound? Does containment hold over each fixed sparse topology, such as a ring with one chord?

Open problem 8.3 (Other arcs and explicit thresholds)**.** Give explicit η_n and $\varepsilon_0(n, t)$ in Theorem 7.4, and decide whether the resetting mechanism breaks containment near every arc of Ω_n whose realising matrix has one-way edges, not only the arc ending at -1.

Declarations

Data and code availability. The accompanying bundle, archived at `https://doi.org/10.5281/zenodo.23687327` and available at `https://github.com/ipitchford/affinity-karpelevich-threshold` (tag `v1.0.0-candidate`), contains the exact verifiers `verify_affine_counterexample.py` (four-state witness), `audit_karpelevich.py`, `audit_all_dimensions.py`, `extension_fingerprint.py` (exact fingerprints of the constructions for $n = 4, 5, 6$ at $t = 1/2$, $\varepsilon = 1/1000$; these identify the matrices and the critical triangle but do not test separation), `certify_higher_dimensions.py` (Theorem 7.6; it also writes the terminal boxes to `exterior_certificates.json`), the separate checker `check_exterior_certificate.py`, and `plot_geometry.py` (figures; illustration only). Each script writes its receipt to an output directory given by `-out` (default `receipts/`) and refuses to overwrite existing files. All scripts were rerun from a clean copy with Python 3.11.14 (`uv run python3 -I`); the verifiers use only

the standard library, and the figure script uses NumPy and Matplotlib. Decimal fields in receipts are for orientation and may differ in the last digits between platforms; all decisive fields are exact.

Literature search. We searched arXiv and the published literature to 27 October 2026 for refutations or restatements of the affine Karpelevich conjecture, including the follow-up works [7, 13, 11, 2]. None contains a counterexample or discusses the conjecture's failure; [7] remarks only that its polygon bound is weaker than the Karpelevich region in the irreversible limit. This is a bounded search, not a priority claim.

Use of AI tools. The research, the computer-assisted certificates, the manuscript text and this revision were produced with the assistance of AI language-model tools, operating under human direction. The tools were used to search the literature, derive and check proofs, write and run the verification scripts, and draft and edit the text. All mathematical claims are supported either by the proofs in the paper or by the exact-arithmetic scripts in the bundle, which can be rerun independently.

Competing interests and funding. None.

A Certificate data for the five- and six-state instances

This appendix prints the data behind Theorem 7.6, so that the certificate can be checked from the paper and the receipt `receipts/higher_dimension_certificates.json`. The terminal boxes (352 for $n = 5$, 578 for $n = 6$) are in `receipts/exterior_certificates.json`; they are too many to print, and `check_exterior_certificate.py` checks them. All numbers below are exact unless marked with a decimal point. Decimal margins are for orientation only and play no part in the proof.

A.1 $n = 5$, $t = 3/5$, $\varepsilon = 1/40000$

Rate matrix. Rows sum to zero and $\alpha = \frac{10001}{10000}$:

$$Q = \frac{1}{200000}\begin{pmatrix} -200020 & 200005 & 5 & 5 & 5 \\ 2 & -200017 & 200005 & 5 & 5 \\ 2 & 120005 & -200017 & 80005 & 5 \\ 2 & 5 & 5 & -200017 & 200005 \\ 80002 & 5 & 5 & 120005 & -200017 \end{pmatrix}.$$

Cycles. The complete graph on 5 states has 37 unoriented simple cycles. The ratio of a cycle c is $K_c = \prod_{(i\to j)\in c} Q_{ij}/Q_{ji}$, oriented so that $K_c \geq 1$; its mean affinity is $\frac{1}{|c|}\log K_c$.

- Dominant: $(1, 2, 5)$, $K = 1600080001$, mean affinity 7.06443982.
- Runner-up: $(1, 2, 3, 4, 5)$, $K = \frac{40966656409612160176001}{576048001}$, mean affinity 6.37906877.

The dominance is checked exactly by $K_c^{|d|} > K_d^{|c|}$. Hence $\beta = (K^{1/3} - 1)/(K^{1/3} + 1)$ with the dominant K; the certificate uses rational bounds with $\beta \in [\beta_{\mathrm{lo}}, \beta_{\mathrm{hi}}]$, $\beta \approx 0.998291513263596$.

Eigenvalue enclosure. A root z^* of $F_{5,t}$ lies within $R = 2 \cdot 10^{-60}$ of a centre z_0 with

$$\operatorname{Re} z_0 = -13649927297835755806022161367548888155557769095807189880822 85 \cdot 2^{-200},$$
$$\operatorname{Im} z_0 = 39817817505677467486305788187873489157099052455782581795 1957 \cdot 2^{-200},$$

so $\lambda = z^* - 1 - V\varepsilon$ is an eigenvalue of Q. The rescaled eigenvalue $\zeta \approx -0.849362124948763 + 0.248186133174560\,\mathrm{i}$ lies in the box $W = [x_{\mathrm{lo}}, x_{\mathrm{hi}}] \times [y_{\mathrm{lo}}, y_{\mathrm{hi}}]$ with

$$\begin{aligned} x_{\mathrm{lo}} &= -2890229542505628556601910027793110259 53 \cdot 2^{-128}, \\ x_{\mathrm{hi}} &= -7225573856264071391504775069482775648 7 \cdot 2^{-126}, \\ y_{\mathrm{lo}} &= 2111334120839863818740555334651506793 9 \cdot 2^{-126}, \\ y_{\mathrm{hi}} &= 4222668241679727637481110669303013587 9 \cdot 2^{-127}, \end{aligned}$$

of width below 1.469×10^{-38}, and $S = \frac{28252422707997040717}{25000000000000000000}$ satisfies $S^2 \min_W |w|^2 \geq 1$.

Farey and arc tests. For each Farey pair the bisection of $[1, S] \times [0, 1]$ visits the listed number of boxes; the terminal boxes cover the rectangle, and on each one the interval extension excludes 0. The last column is the smallest distance from 0 of the excluding real or imaginary enclosure over the terminal boxes. The tightest pairs are $2/5, 1/2$ and its reflection $1/2, 3/5$: the eigenvalue lies just outside the arc between $e^{4\pi\mathrm{i}/5}$ and -1. The radial margin, the distance between $|\zeta|$ and the outermost boundary point on its ray, is $\approx 4.669 \times 10^{-5}$ (floating point, for scale only).

pair	q	s	k	visited	terminal	min. margin
$0, 1/5$	1	5	5	25	13	1.82×10^{-2}
$1/5, 1/4$	4	5	1	7	4	2.57×10^{-2}
$1/4, 1/3$	3	4	1	7	4	7.26×10^{-2}
$1/3, 2/5$	3	5	1	1	1	7.64×10^{-2}
$2/5, 1/2$	2	5	2	307	154	8.78×10^{-7}
$1/2, 3/5$	2	5	2	307	154	8.78×10^{-7}
$3/5, 2/3$	3	5	1	1	1	7.64×10^{-2}
$2/3, 3/4$	3	4	1	7	4	7.26×10^{-2}
$3/4, 4/5$	4	5	1	7	4	2.57×10^{-2}
$4/5, 1$	1	5	5	25	13	1.82×10^{-2}

A.2 $n = 6$, $t = 1/2$, $\varepsilon = 1/8000$

Rate matrix. Rows sum to zero and $\alpha = \frac{1601}{1600}$:

$$Q = \frac{1}{16000} \begin{pmatrix} -16010 & 16002 & 2 & 2 & 2 & 2 \\ 8001 & -16009 & 8002 & 2 & 2 & 2 \\ 1 & 2 & -16009 & 16002 & 2 & 2 \\ 1 & 2 & 8002 & -16009 & 8002 & 2 \\ 1 & 2 & 2 & 2 & -16009 & 16002 \\ 1 & 8002 & 2 & 2 & 8002 & -16009 \end{pmatrix}.$$

Cycles. The complete graph on 6 states has 197 unoriented simple cycles. The ratio of a cycle c is $K_c = \prod_{(i \to j) \in c} Q_{ij}/Q_{ji}$, oriented so that $K_c \geq 1$; its mean affinity is $\frac{1}{|c|} \log K_c$.

- Dominant: $(2, 3, 6)$, $K = 16008001$, mean affinity 5.52953307.
- Runner-up: $(2, 3, 4, 5, 6)$, $K = 256128020001$, mean affinity 5.25378864.

The dominance is checked exactly by $K_c^{|d|} > K_d^{|c|}$. Hence $\beta = (K^{1/3} - 1)/(K^{1/3} + 1)$ with the dominant K; the certificate uses rational bounds with $\beta \in [\beta_{\mathrm{lo}}, \beta_{\mathrm{hi}}]$, $\beta \approx 0.992095680385056$.

Eigenvalue enclosure. A root z^* of $F_{6,t}$ lies within $R = 2 \cdot 10^{-60}$ of a centre z_0 with

$$\begin{aligned} \operatorname{Re} z_0 &= -1405925988550798938004119301164136760896295741542989741518381 \cdot 2^{-200}, \\ \operatorname{Im} z_0 &= 2022362839879629429965876312908275711400889492821858340322 95 \cdot 2^{-199}, \end{aligned}$$

so $\lambda = z^* - 1 - V\varepsilon$ is an eigenvalue of Q. The rescaled eigenvalue $\zeta \approx -0.874425874457829 + 0.253550827099820\,\mathrm{i}$ lies in the box $W = [x_{\mathrm{lo}}, x_{\mathrm{hi}}] \times [y_{\mathrm{lo}}, y_{\mathrm{hi}}]$ with

$$\begin{aligned} x_{\mathrm{lo}} &= -297551706257421405089668952259855843307 \cdot 2^{-128}, \\ x_{\mathrm{hi}} &= -148775853128710702544834476129927921651 \cdot 2^{-127}, \\ y_{\mathrm{lo}} &= 215697188950720580984585307982887713361 \cdot 2^{-126}, \\ y_{\mathrm{hi}} &= 431394377901441161969170615965775427 23 \cdot 2^{-127}, \end{aligned}$$

of width below 1.469×10^{-38}, and $S = \frac{6864780988993527561}{6250000000000000000}$ satisfies $S^2 \min_W |w|^2 \geq 1$.

Farey and arc tests. For each Farey pair the bisection of $[1, S] \times [0, 1]$ visits the listed number of boxes; the terminal boxes cover the rectangle, and on each one the interval extension excludes 0. The last column is the smallest distance from 0 of the excluding real or imaginary enclosure over the terminal boxes. The tightest pairs are $2/5, 1/2$ and its reflection $1/2, 3/5$: the eigenvalue lies just outside the arc between $e^{4\pi\mathrm{i}/5}$ and -1. The radial margin, the distance between $|\zeta|$ and the outermost boundary point on its ray, is $\approx 2.210 \times 10^{-4}$ (floating point, for scale only).

pair	q	s	k	visited	terminal	min. margin
$0, 1/6$	1	6	6	19	10	3.21×10^{-2}
$1/6, 1/5$	5	6	1	3	2	2.67×10^{-1}
$1/5, 1/4$	4	5	1	7	4	1.08×10^{-1}
$1/4, 1/3$	3	4	2	15	8	2.13×10^{-2}
$1/3, 2/5$	3	5	2	7	4	8.34×10^{-2}
$2/5, 1/2$	2	5	3	521	261	4.43×10^{-6}
$1/2, 3/5$	2	5	3	521	261	4.43×10^{-6}
$3/5, 2/3$	3	5	2	7	4	8.34×10^{-2}
$2/3, 3/4$	3	4	2	15	8	2.13×10^{-2}
$3/4, 4/5$	4	5	1	7	4	1.08×10^{-1}
$4/5, 5/6$	5	6	1	3	2	2.67×10^{-1}
$5/6, 1$	1	6	6	19	10	3.21×10^{-2}

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